

Yu Wang

☎ (206) 304-7202 ✉ wangyu7989@gmail.com 💻 yu-wang-200658208 🌐 github.com/759257989

Education

Northeastern University

Sep 2023 - Jan 2026

Master of Science in Computer Science

Relevant Coursework: Algorithms & Data Structures, Distributed Systems, Database Systems, Cloud Computing, Web Development

University of Pennsylvania

Sep 2022 - May 2023

Master of Urban Spatial Analytics

Relevant Coursework: Machine Learning, Statistical & Data Mining Methods, Data Science in Python, Geospatial Cloud Computing

Experience

Software Engineering Intern – Neusoft Corporation, Automotive Innovation Division

Jun 2025 – Sep 2025

- Built a **Java service layer** integrating UI fragments with the **navigation SDK**, improving app stability and increasing crash-free sessions from **92% to 97%**.
- Delivered **10+ production navigation features**, including voice guidance, route preferences, real-time visualization, and UI-SDK state synchronization.
- Refactored legacy modules and core logic, reducing code complexity by **25%** and improving maintainability.
- Resolved high-priority production issues through log analysis, debugging, and cross-functional collaboration with QA and product teams using **Jira**.

Software Engineering Intern – Tech@Crossroads

Dec 2024 – Mar 2025

- Developed **Spring Boot REST APIs for authentication, registration, and event workflow management**, supporting user-facing event platform features from backend implementation to deployment.
- **Optimized MySQL queries and introduced Redis caching**, reducing duplicate queries by 35% and improving p95 latency from 320 ms to 190 ms.
- Built reusable **React and TypeScript** components with Hooks and Axios, reducing duplicate UI code by **25%**.
- Contributed to code reviews, Agile sprint planning, CI/CD integration, and release validation.

Projects

Distributed IoT Job Processing Platform – Go, Kubernetes, Kafka, PostgreSQL, Redis, Prometheus, Grafana

- Built a **distributed platform in Go and Kubernetes**, supporting asynchronous job execution across realtime, standard, and bulk Kafka worker tiers and processing 93,000 jobs in load test.
- **Reduced unexpected 5xx errors from 80% to under 2%** at 1,000 req/s by adding lag-based API backpressure, per-tier overload thresholds, Retry-After responses, and DLQ routing.
- Strengthened reliability with **Redis idempotency, PostgreSQL unique indexes, heartbeat-based recovery**.
- Implemented **observability and load testing** with **Prometheus, Grafana**, burn-rate alerts, k6, and MQTT simulator to validate 1,000 concurrent devices and HPA scaling.

Autonomous Pricing Intelligence Agent – Python, FastAPI, AWS, ChromaDB, LLaMA, BeautifulSoup

- Built an end to end automated pricing system combining **web scraping, RAG, and machine learning prediction**.
- Achieved 0.38 RMSLE and 70% prediction accuracy, deploying as serverless AWS Lambda API for scalable inference.
- Fine-tuned LLaMA models using PEFT quantization, reducing inference cost while maintaining prediction quality.
- **Engineered 400K+ record vector store**, improving retrieval precision by 30% for downstream LLM reasoning.

Feedlyx – Java, Spring Cloud, Redis, Elasticsearch, Docker

- Designed **microservices architecture with Spring Cloud**, improving scalability for high traffic platform.
- Secured **RESTful APIs with JWT authentication**, integrated **API Gateway** for access control and traffic routing.
- Optimized backend query performance using Redis caching and Elasticsearch indexing, reducing database read load by 35% and improving p95 latency from 450 ms to under 180 ms.
- Improved API reliability by containerizing services with Docker and implementing circuit breakers and fallback logic, reducing downtime by 40% and increasing uptime from 99.3% to 99.7%.

Technical Skills

Programming Languages: Java, Python, Go, C++, SQL, TypeScript, JavaScript

Backend Development: Spring Boot, Django, Node.js, Express.js, RESTful APIs, Microservices, Distributed Systems

Frontend Development: React, TypeScript, Tailwind CSS

Databases & Messaging: MySQL, PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ, Elasticsearch

Cloud & DevOps: AWS EC2, RDS, Lambda, Docker, Kubernetes, Helm, CI/CD, GitHub Actions

Testing & Developer Tools: Git, Jira, Postman, JUnit, Pytest, Mockito, Selenium, Apache JMeter, k6, Agile/Scrum